

Question:

The feedback loop in men which regulates the production of sex hormones includes all but which of the following?

Answer Choices:

- A. follicle-stimulating hormone
- B. oxytocin
- C. pituitary gland
- D. hypothalamus
- E. gonadotropin-releasing hormone
- F. luteinizing hormone
- G. testosterone

Reasoning + Answer:

<think> To determine which option does not belong in the feedback loop that regulates the production of sex hormones in men , we need to recall the key components of this feedback loop . This loop involves the hypothalamus , pituitary gland , and the gonads ( testes ). Let's analyze each option :

- \*\* A . Follicle -stimulating hormone ( FSH )\*\* : FSH is a gonadotropin produced by the pituitary gland that stimulates the production of testosterone in men .
- \*\* B . Oxytocin \*\*: This is a hormone that plays a role in social bonding and has various functions in the body , but it is not directly involved in the sex hormone production feedback loop .
- \*\* C . Pituitary gland \*\*: The pituitary gland is a key endocrine gland that produces and releases several hormones , including gonadotropins like FSH and another hormone , luteinizing hormone ( LH ), which are involved in sex hormone production .
- \*\* D . Hypothalamus \*\*: The hypothalamus is part of the brain that regulates various bodily functions , including the sex hormone production feedback loop .
- \*\* E . Gonadotropin -releasing hormone ( GnRH )\*\* : GnRH is produced by the hypothalamus and signals the pituitary gland to release FSH and LH .
- \*\* F . Luteinizing hormone ( LH )\*\* : LH is a gonadotropin produced by the pituitary gland that stimulates the production of testosterone in men .
- \*\* G . Testosterone \*\*: This is a sex hormone and is not a part of the hormone feedback loop .

Oxytocin , being involved in social bonding , is not directly involved in the sex hormone production feedback loop .

<answer> B </answer>